Question ID:191
Topic Name: Mathematics-Section A

Question: The number of points of intersection of $|z-(4+3i)|=2$ and $|z|+|z-4|=6$, $z \in \mathbb{C}$, is

A 0

B 1

C 2

D 3

Answer Given By Candidate:**C**

Question ID:192
Topic Name: Mathematics-Section A

Question: Let $f(x)=\begin{vmatrix} a & -1 & 0 \\ ax & a & -1 \\ ax^2 & ax & a \end{vmatrix}$, $a \in \mathbb{R}$. Then the sum of the squares of all the values of a, for which $2f'(10)-f'(5)+100=0$, is

A 117

B 106

C 125

D 136

Answer Given By Candidate:**C**

Question ID:193
Topic Name: Mathematics-Section A

Question: Let for some real numbers α and β, $a=\alpha-i\beta$. If the system of equations $4ix+(1+i)y=0$ and $8\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right)x+\bar{a}y=0$ has more than one solution, then $\frac{\alpha}{\beta}$ is equal to

A $-2+\sqrt{3}$

B

$2-\sqrt{3}$

C $2+\sqrt{3}$

D $-2-\sqrt{3}$

Answer Given By Candidate:**Not Attempted**

Question ID:194
Topic Name: Mathematics-Section A

Question: Let A and B be two 3×3 matrices such that $AB = I$ and $|A| = \frac{1}{8}$. Then $|adj\,(B\ adj(2A))|$ is equal to

A 16

B 32

C 64

D 128

Answer Given By Candidate:**C**

Question ID:195
Topic Name: Mathematics-Section A

Question: Let $S = 2 + \frac{6}{7} + \frac{12}{7^2} + \frac{20}{7^3} + \frac{30}{7^4} + \ldots\ldots$ Then $4S$ is equal to

A $\left(\frac{7}{3}\right)^2$

B $\frac{7^3}{3^2}$

C $\left(\frac{7}{3}\right)^3$

D $\frac{7^2}{3^3}$

Answer Given By Candidate:**Not Attempted**

Question ID:196
Topic Name: Mathematics-Section A

Question: If a_1, a_2, a_3 ..... and b_1, b_2, b_3..... are A.P., and $a_1 = 2$, $a_{10} = 3$, $a_1 b_1 = 1 = a_{10} b_{10}$, then $a_4\ b_4$ is equal to -

A $\frac{35}{27}$

B 1

C $\frac{27}{28}$

D $\frac{28}{27}$

Answer Given By Candidate:**D**

Question ID:197
Topic Name: Mathematics-Section A

If m and n respectively are the number of local maximum and local minimum points of the function $f(x)=\int_0^{x^2}\frac{t^2-5t+4}{2+e^t}dt$, then the ordered pair (m, n) is

Question: equal to

A $(3, 2)$

B $(2, 3)$

C $(2, 2)$

D $(3, 4)$

Answer Given By Candidate:**B**

Question ID:198
Topic Name: Mathematics-Section A

Let f be a differentiable function in $\left(0, \frac{\pi}{2}\right)$ If $\int_{\cos x}^{1} t^2 f(t)\,dt = \sin^3 x + \cos x$, then $\frac{1}{\sqrt{3}} f'\left(\frac{1}{\sqrt{3}}\right)$ is equal to

Question:

A $6-9\sqrt{2}$

B $6-\frac{9}{\sqrt{2}}$

C $\frac{9}{2}-6\sqrt{2}$

D $\frac{9}{\sqrt{2}}-6$

Answer Given By Candidate:**Not Attempted**

Question ID:199
Topic Name: Mathematics-Section A

The integral $\int_0^1 \frac{1}{7^{\left[\frac{1}{x}\right]}}\,dx$, where $[\cdot]$ denotes the greatest integer function, is equal to

Question:

A $1+6\log_e\left(\frac{6}{7}\right)$

B $1-6\log_e\left(\frac{6}{7}\right)$

C $\log_e\left(\frac{7}{6}\right)$

D $1-7\log_e\left(\frac{6}{7}\right)$

Answer Given By Candidate:**Not Attempted**

Question ID:1910
Topic Name: Mathematics-Section A

If the solution curve of the differential equation $((\tan^{-1} y) - x)\,dy = (1+y^2)\,dx$ passes through the point (1, 0), then the abscissa of the point on the curve whose ordinate is tan(1), is

Question:

A $2e$

B $\frac{2}{e}$

C 2

D $\frac{1}{e}$

Answer Given By Candidate:**Not Attempted**

Question ID:1911
Topic Name: Mathematics-Section A

If the equation of the parabola, whose vertex is at (5, 4) and the directrix is $3x + y - 29 = 0$, is $x^2 + ay^2 + bxy + cx + dy + k = 0$, then $a + b + c + d + k$ is equal to

Question:

A 575

B −575

C 576

D −576

Answer Given By Candidate:**Not Attempted**

Question ID:1912
Topic Name: Mathematics-Section A

Question: The set of values of k, for which the circle $C : 4x^2 + 4y^2 - 12x + 8y + k = 0$ lies inside the fourth quadrant and the point $\left(1, -\frac{1}{3}\right)$ lies on or inside the circle C, is

A an empty set

B $\left(6, \frac{65}{9}\right]$

C $\left[\frac{80}{9}, 10\right)$

D $\left(9, \frac{92}{9}\right]$

Answer Given By Candidate:**D**

Question ID:1913
Topic Name: Mathematics-Section A

Question: Let the foot of the perpendicular from the point (1, 2, 4) on the line $\frac{x+2}{4} = \frac{y-1}{2} = \frac{z+1}{3}$ be P. Then the distance of P from the plane $3x + 4y + 12z + 23 = 0$ is

A 5

B $\frac{50}{13}$

C 4

D $\frac{63}{13}$

Answer Given By Candidate:**A**

Question ID:1914
Topic Name: Mathematics-Section A

Question: The shortest distance between the lines $\frac{x-3}{2} = \frac{y-2}{3} = \frac{z-1}{-1}$ and $\frac{x+3}{2} = \frac{y-6}{1} = \frac{z-5}{3}$, is

A $\frac{18}{\sqrt{5}}$

B $\frac{22}{3\sqrt{5}}$

C $\frac{46}{3\sqrt{5}}$

D $6\sqrt{3}$

Answer Given By Candidate:**A**

Question ID:1915
Topic Name: Mathematics-Section A

Question: Let $\vec{a}$ and $\vec{b}$ be the vectors along the diagonals of a parallelogram having area $2\sqrt{2}$. Let the angle between $\vec{a}$ and $\vec{b}$ be acute, $|\vec{a}|=1$, and $|\vec{a}\cdot\vec{b}|=|\vec{a}\times\vec{b}|$. If $\vec{c}=2\sqrt{2}\left(\vec{a}\times\vec{b}\right)-2\vec{b}$, then an angle between $\vec{b}$ and $\vec{c}$ is

A $\frac{\pi}{4}$

B $-\frac{\pi}{4}$

C $\frac{5\pi}{6}$

D $\frac{3\pi}{4}$

Answer Given By Candidate:**D**

Question ID:1916
Topic Name: Mathematics-Section A

Question: The mean and variance of the data 4, 5, 6, 6, 7, 8, x, y, where $x<y$, are 6 and $\frac{9}{4}$ respectively. Then x^4+y^2 is equal to

A 162

B 320

C 674

D 420

Answer Given By Candidate:**B**

Question ID:1917
Topic Name: Mathematics-Section A

Question: If a point $A(x, y)$ lies in the region bounded by the y-axis, straight lines $2y + x = 6$ and $5x - 6y = 30$, then the probability that $y < 1$ is

A $\frac{1}{6}$

B $\frac{5}{6}$

C

$\frac{2}{3}$

D $\frac{6}{7}$

Answer Given By Candidate:**B**

Question ID:1918
Topic Name: Mathematics-Section A
Question: The value of $\cot\left(\sum_{n=1}^{50}\tan^{-1}\left(\frac{1}{1+n+n^2}\right)\right)$ is

A $\frac{26}{25}$

B $\frac{25}{26}$

C $\frac{50}{51}$

D $\frac{52}{51}$

Answer Given By Candidate:**Not Attempted**

Question ID:1919
Topic Name: Mathematics-Section A
Question: $\alpha = \sin 36°$ is a root of which of the following equation?

A $16x^4 - 10x^2 - 5 = 0$

B $16x^4 + 20x^2 - 5 = 0$

C $16x^4 - 20x^2 + 5 = 0$

D $16x^4 - 10x^2 + 5 = 0$

Answer Given By Candidate:**Not Attempted**

Question ID:1920
Topic Name: Mathematics-Section A
Question: Which of the following statement is a tautology ?

A $((\sim q) \wedge p) \wedge q$

B $((\sim q) \wedge p) \wedge (p \wedge (\sim p))$

C $((\sim q) \wedge p) \vee (p \vee (\sim p))$

D $(p \wedge q) \wedge (\sim(p \wedge q))$

Answer Given By Candidate:**C**

Question ID:1921
Topic Name: Mathematics-Section B

Question: Let $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Define $f : S \to S$ as

$$f(n) = \begin{cases} 2n & , \text{ if } n = 1, 2, 3, 4, 5 \\ 2n - 11 & , \text{ if } n = 6, 7, 8, 9, 10 \end{cases}.$$

Let $g : S \to S$ be a function such that $fog(n) = \begin{cases} n+1 & , \text{ if } n \text{ is odd} \\ n-1 & , \text{ if } n \text{ is even} \end{cases}$,

Then $g(10)\,(g(1) + g(2) + g(3) + g(4) + g(5))$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1922
Topic Name: Mathematics-Section B

Question: Let α, β be the roots of the equation $x^2 - 4\lambda x + 5 = 0$ and α, γ be the roots of the equation $x^2 - \left(3\sqrt{2} + 2\sqrt{3}\right)x + 7 + 3\lambda\sqrt{3} = 0,\ \lambda > 0$. If $\beta + \gamma = 3\sqrt{2}$, then $(\alpha + 2\beta + \gamma)^2$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1923
Topic Name: Mathematics-Section B

Question: Let A be a matrix of order 2×2, whose entries are from the set $\{0, 1, 2, 3, 4, 5\}$. If the sum of all the entries of A is a prime number p, $2 < p < 8$, then the number of such matrices A is _____.

Answer Given By Candidate:**180**

Question ID:1924
Topic Name: Mathematics-Section B

Question: If the sum of the coefficients of all the positive powers of x, in the Binomial expansion of $\left(x^n + \frac{2}{x^5}\right)^7$ is 939, then the sum of all the possible integral values of n is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1925
Topic Name: Mathematics-Section B

Question: Let $[t]$ denote the greatest integer $\le t$ and $\{t\}$ denote the fractional part of t. The integral value of α for which the left hand limit of the function $f(x) = [1 + x] + \frac{\alpha^{2[x]+\{x\}} + [x] - 1}{2[x] + \{x\}}$ at $x = 0$ is equal to $\alpha - \frac{4}{3}$, is _____.

Answer Given By Candidate:**3**

Question ID:1926
Topic Name: Mathematics-Section B

Question: If $y(x) = \left(x^x\right)^x$, $x > 0$, then $\frac{d^2x}{dy^2} + 20$ at $x = 1$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1927
Topic Name: Mathematics-Section B

Question: If the area of the region $\left\{(x, y) : x^{\frac{2}{3}} + y^{\frac{2}{3}} \le 1, x + y \ge 0, y \ge 0\right\}$ is A, then $\frac{256A}{\pi}$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1928
Topic Name: Mathematics-Section B

Question: Let $y = y(x)$ be the solution of the differential equation $\left(1-x^2\right)dy = \left(xy + \left(x^3+2\right)\sqrt{1-x^2}\right)dx$, $-1 < x < 1$, and $y(0) = 0$. If $\int_{\frac{-1}{2}}^{\frac{1}{2}} \sqrt{1-x^2}\, y(x)dx = k$, then k^{-1} is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1929
Topic Name: Mathematics-Section B

Question: Let a circle C of radius 5 lie below the x-axis. The line $L_1 : 4x + 3y + 2 = 0$ passes through the centre P of the circle C and intersects the line $L_2 : 3x - 4y - 11 = 0$ at Q. The line L_2 touches C at the point Q. Then the distance of P from the line $5x - 12y + 51 = 0$ is _____.

Answer Given By Candidate:**1**

Question ID:1930
Topic Name: Mathematics-Section B

Question: Let $S = \{E_1, E_2, \ldots\ldots, E_8\}$ be a sample space of a random experiment such that $P(E_n) = \frac{n}{36}$ for every $n = 1, 2, \ldots\ldots, 8$. Then the number of elements in the set $\left\{A \subseteq S : P(A) \ge \frac{4}{5}\right\}$ is _____.

Answer Given By Candidate:**18**

Question ID:1931
Topic Name: Physics-Section A

Question: The SI unit of a physical quantity is pascal-second. The dimensional formula of this quantity will be :

A $[ML^{-1}T^{-1}]$

B $[ML^{-1}T^{-2}]$

C $[ML^{2}T^{-1}]$

D $[M^{-1}L^{3}T^{0}]$

Answer Given By Candidate:**A**

Question ID:1932
Topic Name: Physics-Section A

Question: The distance of the Sun from earth is 1.5×10^{11} m and its angular diameter is (2000) s when observed from the earth. The diameter of the Sun will be :

A 2.45×10^{10} m

B 1.45×10^{10} m

C 1.45×10^{9} m

D 0.14×10^{9} m

Answer Given By Candidate:**C**

Question ID:1933
Topic Name: Physics-Section A

Question: When a ball is dropped into a lake from a height 4.9 m above the water level, it hits the water with a velocity v and then sinks to the bottom with the constant velocity v. It reaches the bottom of the lake 4.0 s after it is dropped. The approximate depth of the lake is :

A 19.6 m

B 29.4 m

C 39.2 m

D 73.5 m

Answer Given By Candidate:**C**

Question ID:1934
Topic Name: Physics-Section A

Question: One end of a massless spring of spring constant k and natural length l_0 is fixed while the other end is connected to a small object of mass m lying on a frictionless table. The spring remains horizontal on the table. If the object is made to rotate at an angular velocity ω about an axis passing trough fixed end, then the elongation of the spring will be :

A $\frac{k - m\omega^2 l_0}{m\omega^2}$

B $\frac{m\omega^2 l_0}{k + m\omega^2}$

C $\frac{m\omega^2 l_0}{k - m\omega^2}$

D $\frac{k + m\omega^2 l_0}{m\omega^2}$

Answer Given By Candidate:**C**

Question ID:1935
Topic Name: Physics-Section A

Question: A stone tide to a string of length L is whirled in a vertical circle with the other end of the string at the centre. At a certain instant of time, the stone is at its lowest position and has a speed u. The magnitude of change in its velocity, as it reaches a position where the string is horizontal, is $\sqrt{x(u^2 - gL)}$. The value of x is-

A 3

B 2

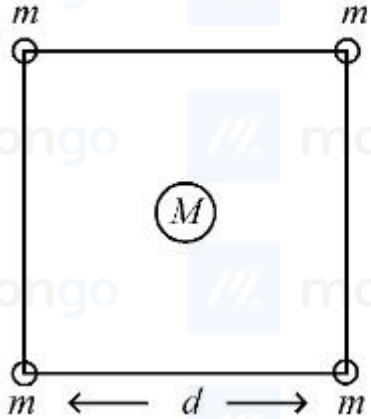
C 1

D 5

Answer Given By Candidate:**B**

Question ID:1936
Topic Name: Physics-Section A

Question: Four spheres each of mass m form a square of side d (as shown in figure). A fifth sphere of mass M is situated at the centre of square. The total gravitational potential energy of the system is :



A $-\frac{Gm}{d}\left[(4+\sqrt{2})m + 4\sqrt{2}M\right]$

B

$-\frac{Gm}{d}\left[(4+\sqrt{2})M+4\sqrt{2}m\right]$

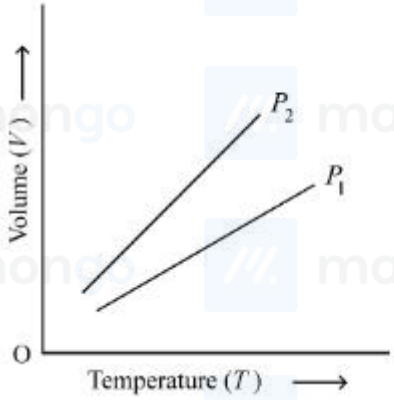
C $-\frac{Gm}{d}\left[3m^2+4\sqrt{2}M\right]$

D $-\frac{Gm}{d}\left[6m^2+4\sqrt{2}M\right]$

Answer Given By Candidate:**A**

Question ID:1937
Topic Name: Physics-Section A

For a perfect gas, two pressures P_1 and P_2 are shown in figure. The graph shows :



Question:

A $P_1 > P_2$

B $P_1 < P_2$

C $P_1 = P_2$

D Insufficient data to draw any conclusion

Answer Given By Candidate:**A**

Question ID:1938
Topic Name: Physics-Section A

According to kinetic theory of gases,
A. The motion of the gas molecules freezes at 0°C.
B. The mean free path of gas molecules decreases if the density of molecules is increased.
C. The mean free path of gas molecules increases if temperature is increased keeping pressure constant.
D. Average kinetic energy per molecule per degree of freedom is $\frac{3}{2}k_BT$ (for monoatomic gases).

Question: Choose the most appropriate answer from the options given below:

A A and C only

B B and C only

C A and B only

D C and D only

Answer Given By Candidate:**B**

Question ID:1939
Topic Name: Physics-Section A

Question: A lead bullet penetrates into a solid object and melts. Assuming that 40% of its kinetic energy is used to heat it, the initial speed of bullet is :
(Given, initial temperature of the bullet = 127°C,
Melting point of the bullet = 327°C,
Latent heat of fusion of lead = 2.5×10^4 J kg^{-1},
Specific heat capacity of lead = 125 J/kg K)

A 125 ms^{-1}

B 500 ms^{-1}

C 250 ms^{-1}

D 600 ms^{-1}

Answer Given By Candidate:**B**

Question ID:1940
Topic Name: Physics-Section A

Question: The equation of a particle executing simple harmonic motion is given by $x = \sin \pi \left(t + \frac{1}{3} \right) m$. At t = 1s, the speed of particle will be
(Given: $\pi = 3.14$) -

A 0 cm s^{-1}

B 157 cm s^{-1}

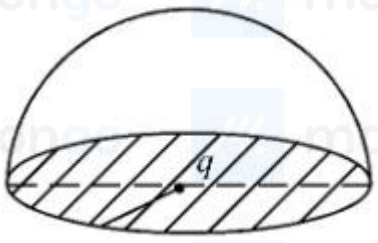
C 272 cm s^{-1}

D 314 cm s^{-1}

Answer Given By Candidate:**B**

Question ID:1941
Topic Name: Physics-Section A

Question: If a charge q is placed at the centre of a closed hemispherical non-conducting surface, the total flux passing through the flat surface would be :



A

$\frac{q}{\epsilon_0}$

B $\frac{q}{2\epsilon_0}$

C $\frac{q}{4\epsilon_0}$

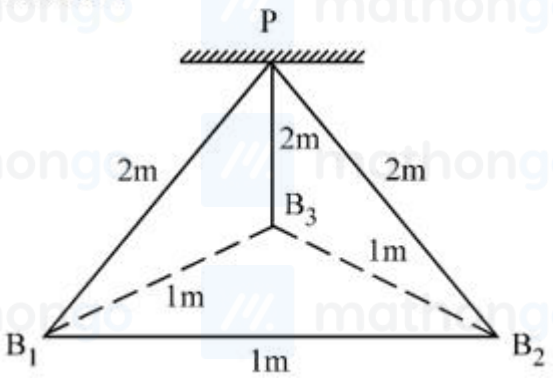
D $\frac{q}{2\pi\epsilon_0}$

Answer Given By Candidate:**B**

Question ID:1942
Topic Name: Physics-Section A

Question:

Three identical charged balls each of charge 2 C are suspended from a common point P by silk threads of 2 m each (as shown in figure). They form an equilateral triangle of side 1m.
The ratio of net force on a charged ball to the force between any two charged balls will be :



A 1:1

B 1:4

C $\sqrt{3}:2$

D $\sqrt{3}:1$

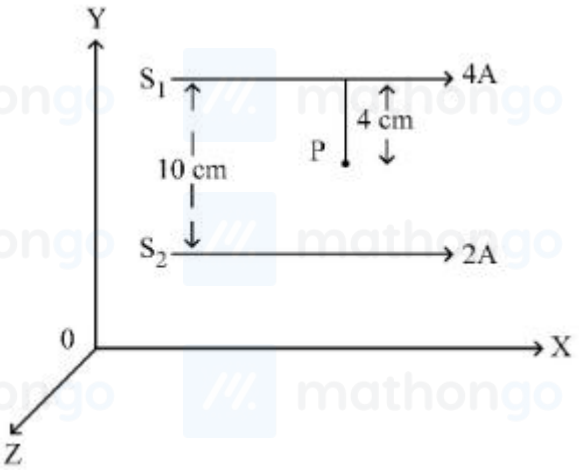
Answer Given By Candidate:**D**

Question ID:1943
Topic Name: Physics-Section A

Two long parallel conductors S_1 and S_2 are separated by a distance 10 cm and carrying currents of 4A and 2A respectively. The conductors are placed along x – axis in X-Y plane. There is a point P located between the conductors (as shown in figure).

A charge particle of 3π coulomb is passing through the point P with velocity $\vec{v} = (2\hat{i} + 3\hat{j})$ m/s; where $\hat{i}$ & $\hat{j}$ represents unit vector along x & y axis respectively.

The force acting on the charge particle is $4\pi \times 10^{-5}(-x\hat{i} + 2\hat{j})\ N$. The value of x is :



Question:

A 2

B 1

C 3

D -3

Answer Given By Candidate:**C**

Question ID:1944
Topic Name: Physics-Section A

Question: If L, C and R are the self inductance, capacitance and resistance respectively, which of the following does not have the dimension of time?

A RC

B $\frac{L}{R}$

C $\sqrt{LC}$

D $\frac{L}{C}$

Answer Given By Candidate:**D**

Question ID:1945
Topic Name: Physics-Section A

Given below are two statements :
Statement I: A time varying electric field is a source of changing magnetic field and vice-versa. Thus a disturbance in electric or magnetic field creates EM waves.
Statement II: In a material medium, the EM wave travels with speed $v = \frac{1}{\sqrt{\mu_0 \in_0}}$.
In the light of the above statements, choose the correct answer from the options given below.

Question:

A Both statement I and statement II are true

B Both statement I and statement II are false

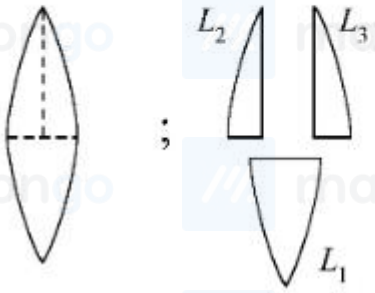
C Statement I is correct but statement II is false

D Statement I is incorrect but statement II is true

Answer Given By Candidate:**C**

Question ID:1946
Topic Name: Physics-Section A

A convex lens has power P. It is cut into two halves along its principal axis. Further one piece (out of the two halves) is cut into two halves perpendicular to the principal axis (as shown in figures). Choose the incorrect option for the reported pieces.



Question:

A Power of $L_1 = \frac{P}{2}$

B Power of $L_2 = \frac{P}{2}$

C Power of $L_3 = \frac{P}{2}$

D Power of $L_1 = P$

Answer Given By Candidate:**A**

Question ID:1947
Topic Name: Physics-Section A
Question: If a wave gets refracted into a denser medium, then which of the following is true?

A wavelength, speed and frequency decreases.

B wavelength increases, speed decreases and frequency remains constant.

C wavelength and speed decreases but frequency remains constant.

D wavelength, speed and frequency increases.

Answer Given By Candidate:**C**

Question ID:1948
Topic Name: Physics-Section A

Question: Given below are two statements:
Statement I: In hydrogen atom, the frequency of radiation emitted when an electron jumps from lower energy orbit (E_1) to higher energy orbit (E_2), is given as $hf = E_1 - E_2$
Statement II: The jumping of electron from higher energy orbit (E_2) to lower energy orbit (E_1) is associated with frequency of radiation given as $f = (E_2 - E_1)/h$
This condition is Bohr's frequency condition.
In the light of the above statements, choose the correct answer from the options given below:

A Both statement I and statement II are true.

B Both statement I and statement II are false.

C Statement I is correct but statement II is false.

D Statement I is incorrect but statement II is true.

Answer Given By Candidate:**D**

Question ID:1949
Topic Name: Physics-Section A

Question: For a transistor to act as a switch, it must be operated in

A Active region.

B Saturation state only.

C Cut-off state only.

D Saturation and cut-off state.

Answer Given By Candidate:**Not Attempted**

Question ID:1950
Topic Name: Physics-Section A

We do not transmit low frequency signal to long distances because-
(a) The size of the antenna should be comparable to signal wavelength which is unreal solution for a signal of longer wavelength.
(b) Effective power radiated by a long wavelength baseband signal would be high.
(c) We want to avoid mixing up signals transmitted by different transmitter simultaneously.
(d) Low frequency signal can be sent to long distances by superimposing with a high frequency wave as well.

Question: Therefore, the most suitable option will be :

A All statements are true

B (a), (b) and (c) are true only

C (a), (c) and (d) are true only

D (b), (c) and (d) are true only

Answer Given By Candidate:**C**

Question ID:1951
Topic Name: Physics-Section B

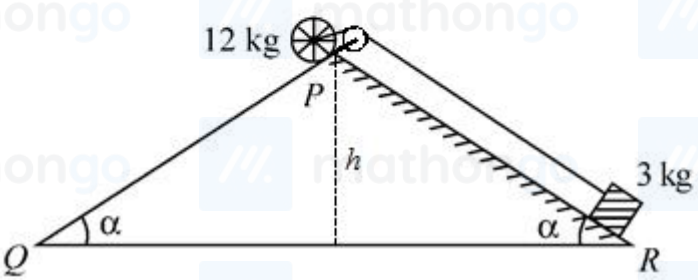
A mass of 10 kg is suspended vertically by a rope of length 5 m from the roof. A force of 30 N is applied at the middle point of rope in horizontal direction. The angle made by upper half of the rope with vertical is $\theta = \tan^{-1}(x \times 10^{-1})$. The value of x is _____.

Question: (Given, $g = 10\ m/s^2$)

Answer Given By Candidate:**Not Attempted**

Question ID:1952
Topic Name: Physics-Section B

A rolling wheel of 12 kg is on an inclined plane at position P and connected to a mass of 3 kg through a string of fixed length and pulley as shown in figure.
Consider PR as friction free surface.
The velocity of centre of mass of the wheel when it reaches at the bottom Q of the inclined plane PQ will be $\frac{1}{2}\sqrt{xgh}\ m/s$. The value of x is _____.



Question:

Answer Given By Candidate:**Not Attempted**

Question ID:1953
Topic Name: Physics-Section B

Question: A diatomic gas ($\gamma = 1.4$) does 400J of work when it is expanded isobarically. The heat given to the gas in the process is _____ J.

Answer Given By Candidate: **Not Attempted**

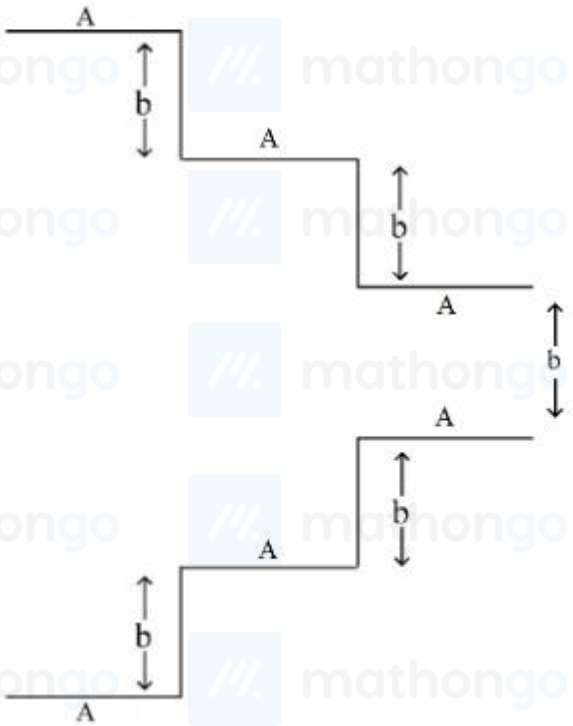
Question ID: 1954
Topic Name: Physics-Section B

Question: A particle executes simple harmonic motion. Its amplitude is 8 cm and time period is 6 s. The time it will take to travel from its position of maximum displacement to the point corresponding to half of its amplitude, is _____ s.

Answer Given By Candidate: **1**

Question ID: 1955
Topic Name: Physics-Section B

Question: A parallel plate capacitor is made up of stair like structure with a plate area A of each stair and that is connected with a wire of length b, as shown in the figure. The capacitance of the arrangement is $\frac{x}{15}\frac{\epsilon_0 A}{b}$, The value of x is _____.



Answer Given By Candidate: **23**

Question ID: 1956
Topic Name: Physics-Section B

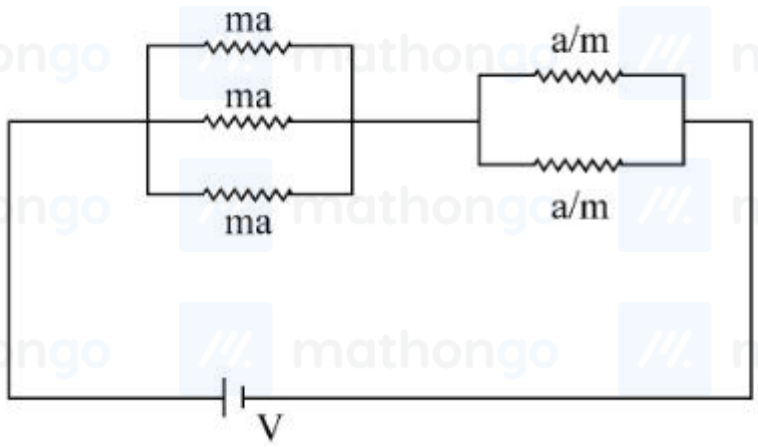
Question: The current density in a cylindrical wire of radius $r = 4.0$ mm is 1.0×10^6 A/m^2. The current through the outer portion of the wire between radial distances $\frac{r}{2}$ and r is $x\pi$ A; where x is _____.

Answer Given By Candidate:12

Question ID:1957
Topic Name: Physics-Section B

Question:

In the given circuit 'a' is an arbitrary constant. The value of m for which the equivalent circuit resistance is minimum, will be $\sqrt{\frac{x}{2}}$. The value of x is _____.



Answer Given By Candidate:Not Attempted

Question ID:1958
Topic Name: Physics-Section B

Question:

A deuteron and a proton moving with equal kinetic energy enter into to a uniform magnetic field at right angle to the field. If r_d and r_p are the radii of their circular paths respectively, then the ratio $\frac{r_d}{r_p}$ will be $\sqrt{x}:1$ where x is _____.

Answer Given By Candidate:2

Question ID:1959
Topic Name: Physics-Section B

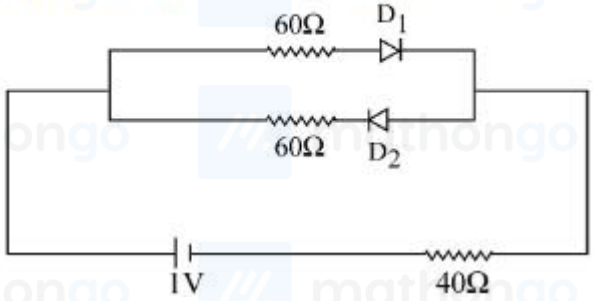
Question:

A metallic rod of length 20 cm is placed in North-South direction and is moved at a constant speed of 20 m/s towards East. The horizontal component of the Earth's magnetic field at that place is 4×10^{-3} T and the angle of dip is 45°. The emf induced in the rod is _____ mV.

Answer Given By Candidate:Not Attempted

Question ID:1960
Topic Name: Physics-Section B

Question: The cut-off voltage of the diodes (shown in figure) in forward bias is 0.6 V. The current through the resister of 40 Ω is _____ mA.

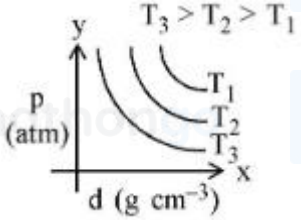


Answer Given By Candidate:**4**

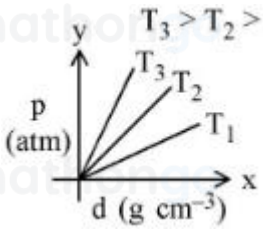
Question ID:1961
Topic Name: Chemistry-Section A

Question: Which amongst the given plots is the correct plot for pressure (p) vs density (d) for an ideal gas?

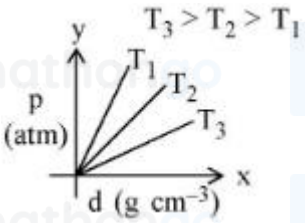
A



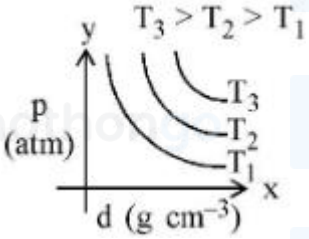
B



C



D



Answer Given By Candidate:**B**

Question ID:1962
Topic Name: Chemistry-Section A

Question: Identify the **incorrect** statement for PCl_5 from the following.

A In this molecule, orbitals of phosphorous are assumed to undergo sp^3d hybridization.

B The geometry of PCl_5 is trigonal bipyramidal.

C PCl_5 has two axial bonds stronger than three equatorial bonds.

D The three equatorial bonds of PCl_5 lie in a plane.

Answer Given By Candidate:**C**

Question ID:1963
Topic Name: Chemistry-Section A
Question:
Statement I: Leaching of gold with cyanide ion in absence of air / O_2 leads to cyano complex of Au(III).
Statement II: Zinc is oxidized during the displacement reaction carried out for gold extraction.
In the light of the above statements, choose the **correct** answer from the options given below.

A Both statement I and statement II are correct.

B Both statement I and statement II are incorrect.

C Statement I is correct but statement II is incorrect.

D Statement I is incorrect but statement II is correct.

Answer Given By Candidate:**A**

Question ID:1964
Topic Name: Chemistry-Section A
Question: The correct order of increasing intermolecular hydrogen bond strength is

A $HCN < H_2O < NH_3$

B $HCN < CH_4 < NH_3$

C $CH_4 < HCN < NH_3$

D $CH_4 < NH_3 < HCN$

Answer Given By Candidate:**C**

Question ID:1965
Topic Name: Chemistry-Section A
Question: The correct order of increasing ionic radii is

A $Mg^{2+} < Na^{+} < F^{-} < O^{2-} < N^{3-}$

B $N^{3-} < O^{2-} < F^{-} < Na^{+} < Mg^{2+}$

C $F^{-} < Na^{+} < O^{2-} < Mg^{2+} < N^{3-}$

D $Na^{+} < F^{-} < Mg^{2+} < O^{2-} < N^{3-}$

Answer Given By Candidate:**A**

Question ID:1966
Topic Name: Chemistry-Section A
Question: The gas produced by treating an aqueous solution of ammonium chloride with sodium nitrite is

A NH_3

B N_2

C N_2O

D Cl_2

Answer Given By Candidate:**A**

Question ID:1967
Topic Name: Chemistry-Section A
Question: Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.
Assertion A: Flourine forms one oxoacid.
Reason R: Flourine has smallest size amongst all halogens and is highly electronegative.
In the light of the above statements, choose the ***most appropriate*** answer from the options given below.

A Both **A** and **R** are correct and **R** is the correct explanation of **A.**

B Both **A** and **R** are correct but **R** is NOT the correct explanation of **A.**

C **A** is correct but **R** is not correct.

D **A** is not correct but **R** is correct.

Answer Given By Candidate:**D**

Question ID:1968
Topic Name: Chemistry-Section A
Question: In 3d series, the metal having the highest M^{2+}/M standard electrode potential is

A Cr

B Fe

C Cu

D Zn

Answer Given By Candidate:**C**

Question ID:1969
Topic Name: Chemistry-Section A

The 'f' orbitals are half and completely filled, respectively in lanthanide ions

Question: [Given : Atomic no. Eu, 63; Sm, 62; Tm, 69; Tb, 65; Yb, 70; Dy, 66]

A Eu^{2+} and Tm^{2+}

B Sm^{2+} and Tm^{3+}

C Tb^{4+} and Yb^{2+}

D Dy^{3+} and Yb^{3+}

Answer Given By Candidate:**C**

Question ID:1970
Topic Name: Chemistry-Section A

Arrange the following coordination compounds in the increasing order of magnetic moments. (Atomic numbers: Mn = 25; Fe = 26)

A. $[FeF_6]^{3-}$
B. $[Fe(CN)_6]^{3-}$
C. $[MnCl_6]^{3-}$ (high spin)
D. $[Mn(CN)_6]^{3-}$

Question: Choose the correct answer from the options given below:

A A < B < D < C

B B < D < C < A

C A < C < D < B

D B < D < A < C

Answer Given By Candidate:**B**

Question ID:1971
Topic Name: Chemistry-Section A

On the surface of polar stratospheric clouds, hydrolysis of chlorine nitrate gives A and B while its reaction with HCl produces B and C. A, B and C are, respectively

Question:

A HOCl, HNO_3, Cl_2

B Cl_2, HNO_3, HOCl

C $HClO_2$, HNO_2, HOCl

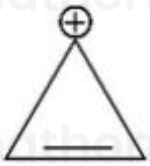
D HOCl, HNO_2, Cl_2O

Answer Given By Candidate:**A**

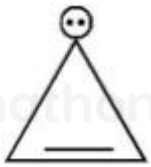
Question ID:1972
Topic Name: Chemistry-Section A

Question: Which of the following is most stable?

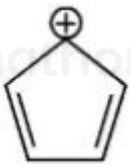
A



B



C



D

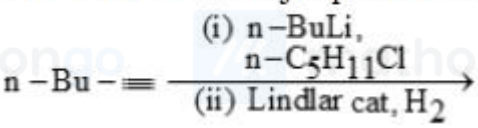


Answer Given By Candidate:**A**

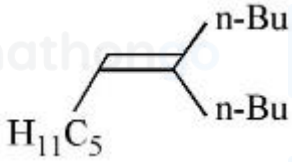
Question ID:1973

Topic Name: Chemistry-Section A

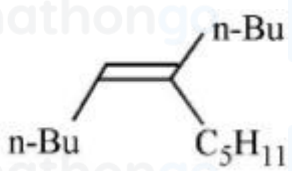
Question: What will be the major product of following sequence of reactions?



A



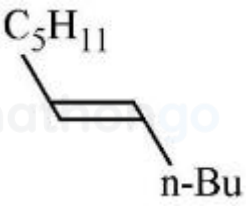
B



C



D

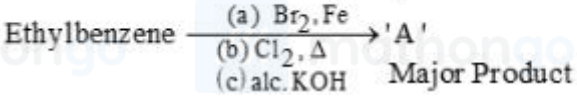


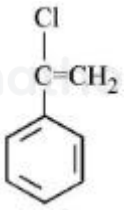
Answer Given By Candidate:**Not Attempted**

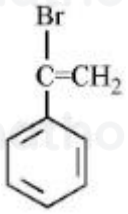
Question ID:1974
Topic Name: Chemistry-Section A

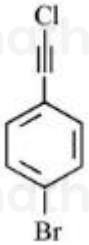
Question:

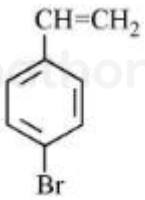
Product 'A' of following sequence of reactions is



A


B


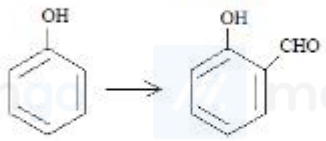
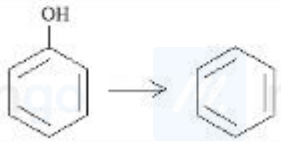
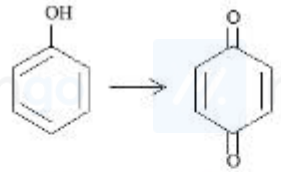
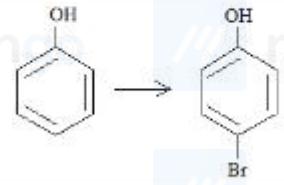
C


D


Answer Given By Candidate:**D**

Question ID:1975
Topic Name: Chemistry-Section A

Match **List I** with **List II.**

List I	List II
A. 	I. Br_2 in CS_2
B. 	II. $Na_2Cr_2O_7/H_2SO_4$
C. 	III. Zn
D. 	IV. $CHCl_3/NaOH$

Question: Choose the correct answer from the options given below:

A A-IV, B-III, C-II, D-I

B A-IV, B-III, C-I, D-II

C A-II, B-III, C-I, D-IV

D A-IV, B-II, C-III, D-I

Answer Given By Candidate:**A**

Question ID:1976
Topic Name: Chemistry-Section A

Question: Decarboxylation of all six possible forms of diaminobenzoic acids $C_6H_3(NH_2)_2COOH$ yields three products A, B and C. Three acids give a product 'A', two acids gives a product 'B' and one acid give a product 'C'. The melting point of product 'C' is

A 63°C

B 90°C

C 104°C

D 142°C

Answer Given By Candidate:**Not Attempted**

Question ID:1977
Topic Name: Chemistry-Section A
Question: Which is true about Buna-N?

A It is a linear polymer of 1,3-butadiene.

B It is obtained by copolymerization of 1,3-butadiene and styrene.

C It is obtained by copolymerization of 1,3-butadiene and acrylonitrile.

D The suffix N in Buna-N stands for its natural occurrence.

Answer Given By Candidate:**Not Attempted**

Question ID:1978
Topic Name: Chemistry-Section A

Question: Given below are two statements.
Statement I: Maltose has two α-D-glucose units linked at C_1 and C_4 and is a reducing sugar.
Statement II: Maltose has two monosaccharides: α-D-glucose and β-D-glucose linked at C_1 and C_6 and it is a non-reducing sugar.
In the light of the above statements, choose the **correct** answer from the options given below.

A Both Statement I and Statement II are true.

B Both Statement I and Statement II are false.

C Statement I is true but Statement II is false.

D Statement I is false but Statement II is true.

Answer Given By Candidate:**Not Attempted**

Question ID:1979
Topic Name: Chemistry-Section A

Question: Match **List I** with **List II**.

List I	List II
A. Antipyretic	I. Reduces pain
B. Analgesic	II. Reduces stress
C. Tranquilizer	III. Reduces fever
D. Antacid	IV. Reduces acidity(stomach)

Question: Choose the correct answer from the options given below:

A A-III, B-I, C-II, D-IV

B A-III, B-I, C-IV, D-II

C A-I, B-IV, C-II, D-III

D A-I, B-III, C-II, D-IV

Answer Given By Candidate:**A**

Question ID:1980
Topic Name: Chemistry-Section A

Match **List I** with **List II**.

List I (Anion)	List II (gas evolved on reaction with dil.H_2SO_4)
A. CO_3^{2-}	I. Colourless gas which turns lead acetate paper black.
B. S^{2-}	II. Colourless gas which turns acidified potassium dichromate solution green.
C. SO_3^{2-}	III. Brown fumes which turns acidified KI solution containing starch blue.
D. NO_2^-	IV. Colourless gas evolved with brisk effervescence, which turns lime water milky.

Question: Choose the correct answer from the options given below:

A A-III, B-I, C-II, D-IV

B A-II, B-I, C-IV, D-III

C A-IV, B-I, C-III, D-II

D A-IV, B-I, C-II, D-III

Answer Given By Candidate:**D**

Question ID:1981
Topic Name: Chemistry-Section B

116 g of a substance upon dissociation reaction, yields 7.5 g of hydrogen, 60 g of oxygen and 48.5 g of carbon. Given that the atomic masses of H, O and C are 1, 16 and 12, respectively. The data agrees with how many formulae of the following ?

_____.

Question: A. CH_3COOH B. HCHO C. CH_3OOCH_3 D. CH_3CHO

Answer Given By Candidate:**Not Attempted**

Question ID:1982
Topic Name: Chemistry-Section B

Consider the following set of quantum numbers.

	n	l	m_l
A.	3	3	−3
B.	3	2	−2
C.	2	1	+1
D.	2	2	+2

Question: The number of correct sets of quantum numbers is _____.

Answer Given By Candidate:**2**

Question ID:1983
Topic Name: Chemistry-Section B

Question: BeO reacts with HF in presence of ammonia to give [A] which on thermal decomposition produces [B] and ammonium fluoride. Oxidation state of Be in [A] is _____

Answer Given By Candidate:**Not Attempted**

Question ID:1984
Topic Name: Chemistry-Section B

Question: When 5 moles of He gas expand isothermally and reversibly at 300 K from 10 litre to 20 litre, the magnitude of the maximum work obtained is _____J. [nearest integer](Given : R = 8.3 J $K^{-1}mol^{-1}$ and log 2 = 0.3010)

Answer Given By Candidate:**8619**

Question ID:1985
Topic Name: Chemistry-Section B

Question: A solution containing 2.5×10^{-3} kg of a solute dissolved in 75×10^{-3} kg of water boils at 373.535 K. The molar mass of the solute is _____ g mol^{-1}. [nearest integer] (Given : $K_b(H_2O)$ = 0.52 K kg mol^{-1} and boiling point of water = 373.15 K)

Answer Given By Candidate:**Not Attempted**

Question ID:1986
Topic Name: Chemistry-Section B

Question: pH value of 0.001 M NaOH solution is _____.

Answer Given By Candidate:**11**

Question ID:1987
Topic Name: Chemistry-Section B

For the reaction taking place in the cell:

$Pt(s)|H_2(g)|H^+(aq)||Ag^+(aq)|Ag(s)$

$E^\circ_{cell} = +0.5332$ V.

Question: The value of $\Delta_f G^\ominus$ is _____ kJ mol^{-1}. (in nearest integer)

Answer Given By Candidate: **Not Attempted**

Question ID: 1988

Topic Name: Chemistry-Section B

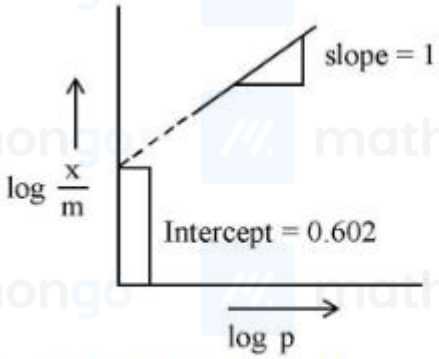
It has been found that for a chemical reaction with rise in temperature by 9 K the rate constant gets doubled. Assuming a reaction to be occurring at 300 K, the value of activation energy is found to be _____kJ mol^{-1}. [nearest integer]

Question: (Given ln10 = 2.3, R = 8.3 J K^{-1} mol^{-1}, log 2 = 0.30)

Answer Given By Candidate: **59**

Question ID: 1989

Topic Name: Chemistry-Section B



Question: If the initial pressure of a gas is 0.03 atm, the mass of the gas adsorbed per gram of the adsorbent is _____ $\times 10^{-2}$g.

Answer Given By Candidate: **Not Attempted**

Question ID: 1990

Topic Name: Chemistry-Section B

0.25 g of an organic compound containing chlorine gave 0.40 g of silver chloride in Carius estimation. The percentage of chlorine present in the compound is _____. [in nearest integer]

Question: (Given: Molar mass of Ag is 108 g mol^{-1} and that of Cl is 35.5 g mol^{-1})

Answer Given By Candidate: **40**